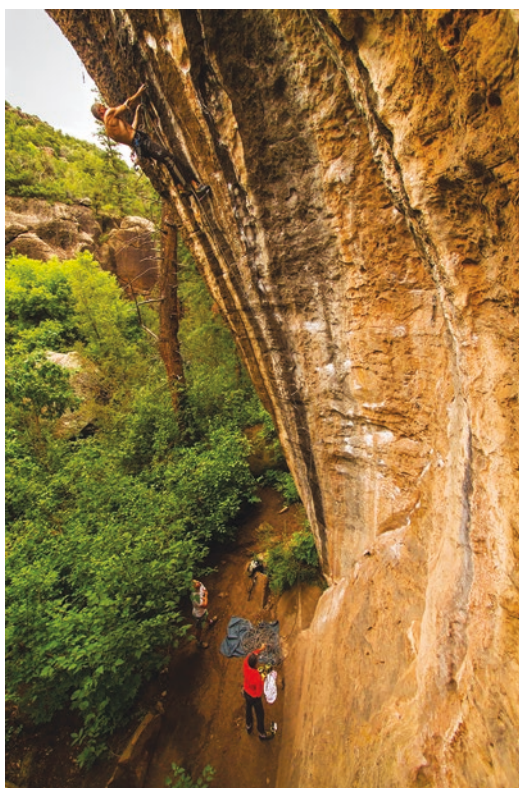


Climbing in Older Athletes

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Volker Schöffl in Mill Creek Canyon, Moab, Utah, USA,
Photo Michael Simon

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15.1 Introduction

Physical activity has been established to be one of the most influential contributing factors to remaining healthy in later stages of life. While many population structures suffer from obesity, immobilization, and improper nutritional behavior, many older adults are very active in sports, some even in the higher age ranges of the population. Rock climbing is one of the sports that are well known to engage athletes on a life-long basis. The sport has proven to have health-promoting effects and an overall low injury hazard in the past [1–4]. However, modern climbing and training methods have changed, and a wider spectrum of climbing-related injuries can now be detected [5]. Several studies on climbing injuries have been conducted in the past, focusing on injury grading and incidences as well as on diagnostic and treatment strategies [6]. However, injury surveillance studies for rock climbing injuries in older athletes are still rare [7]. Furthermore, the rapid increase of athletes in today’s generation of climbers aged between 20 and 30 will likely lead to a wave of elder athletes in the future.

Available studies underlined the fact that overall climbing injury incidence is minor and that most injuries are of a minor grade [4, 6]. Acute injuries in climbing are related to a fall onto the lower leg or from performing a hard move and injuring the upper extremities [3, 8]. Beyond this, overuse injuries can result from permanent undue stress and overload, mainly to the upper extremi-

ties. Most climbing-related scientific studies tend to focus on sport-specific injuries such as flexor tendon pulley injuries, since these injuries occur the most frequently [6, 9]. Expecting a rising number of very active, older rock climbing athletes due to the increase in popularity of climbing and bouldering, it is important to further analyze this patient collective [8].

15.2 Acute and Overuse Injuries

A recently published study investigated 198 patients ≥ 35 years of age (150 [76%] male, 48 [24%] female) with a total number of 275 independent injuries caused by rock climbing or bouldering [10]. Among the 275 injuries, 187 (68%) were overuse injuries and 88 (32%) were acute injuries. The percentage of bouldering in the climbing routine and training was stated by the athletes as 44.5 ± 6.6 (0–100). None of the athletes in the age group >65 years of age reported partaking bouldering on a regular basis. Representing 15% of all injuries, a finger pulley laceration (single or multiple pulley rupture) was the most frequent diagnosis, followed by finger joint capsulitis (12%), subacromial impingement (10%), and finger tenosynovitis (7%) (Table 15.1). With a ratio of three to one, a significantly higher rate of injured male climbers was found compared to injured female climbers. Climbing levels and climbing experience were both significantly higher in males. Thereby, sex did not influence the severity of injury. These findings are in accor-

Table 15.1 Ten most frequent injuries in the older rock climbing athlete [10]

Injuries	All (n = 275)	Age 35–49 (n = 217)	Age 50–64 (n = 54)	Age ≥ 65 (n = 4)
Pulley injury	41 (15%)	33 (15%)	8 (15%)	None
Capsulitis (finger)	34 (12%)	27 (12%)	6 (11%)	1 (25%)
Impingement (shoulder)	28 (10%)	19 (8%)	8 (15%)	1 (25%)
Tenosynovitis	20 (7%)	19 (8%)	1 (2%)	None
SLAP tear	19 (6%)	17 (7%)	2 (4%)	None
Epicondylitis	15 (6%)	13 (5%)	2 (4%)	None
Osteoarthritis (finger)	12 (4%)	8 (4%)	4 (7%)	None
Dupuytren disease	9 (3%)	6 (3%)	3 (6%)	None
Strain finger joint capsule	9 (3%)	8 (4%)	1 (2%)	None
Strain finger flexor tendon	8 (3%)	7 (3%)	1 (2%)	None

SLAP superior labral tear from anterior to posterior

dance with most other available data [3, 4, 9]. The average UIAA injury score was overall low (2.0 ± 0.3 [1–4]). None of the athletes suffered a UIAA grade 5 injury, and none of them died (UIAA 6). Climbing level (UIAA level, $p < 0.001$) and climbing experience (climbing years, $p < 0.001$) were both significantly higher in men than in women. Against the findings of two previously published studies, we could not confirm a correlation between the climbing difficulty (UIAA score) and the development of overuse injuries [3, 4]. This finding is remarkable, as one would expect overuse injuries to be most frequent in very old athletes with decades of climbing history. No significant association between the climber’s UIAA climbing level, climbing hours/week, years of climbing, height, weight, and sex and the development of an acute or overuse injury was found. The UIAA injury score was also not significantly influenced by any of these individual parameters.

Athletes affected by acute injuries did not have significantly higher UIAA scores than those affected by overuse injuries. As reported in investigations of younger rock climbing athletes, the vast majority of injuries in the study (90%) affected the upper extremities [2, 6, 9]. In a prospective analysis of more than half a million entries to an indoor climbing and bouldering gym, the authors found an injury rate of 0.02/1000 h, of which only 15 were of UIAA injury scale grade 2, 13 UIAA 3, and 2 UIAA 4

[11]. No higher graded injuries (UIAA injury scale) were reported, and no fatalities occurred in the study. The severity of injuries in older athletes is also mostly low, with a UIAA score of 2 (defined as “moderate severe injury,...”) in 93% of all cases [12]. Only six acute cases were classified as UIAA grade 3 injuries, and one poly-traumatic patient was classified as a UIAA grade 4 injury. This overall minor grade of injury severity is mostly in accordance with the previously published data. It is notable that there is a slight increase of more severe injuries graded UIAA 3 and higher compared to prior studies on younger athletes [3, 4]. Contrary to expectations that more severe injuries would be more probable in younger, more aggressive, and occasionally reckless athletes, the study might confirm the recent trend in which we have seen an overall shift into more serious injuries over the last 5 years [5]. This finding is most likely caused by indoor bouldering and the rising number of people being enthusiastic about this sport.

15.3 Degenerative Conditions

Among overuse injuries, degenerative conditions were defined as non-reversible injuries that had worsened over time. Subgroup analysis showed a weak correlation of the climber’s age with the development of a degenerative condition (Tables 15.1 and 15.2). Among the degenerative condi-

Table 15.2 Most frequent degenerative conditions in the older rock climbing athlete [10]

Injuries	All ($n = 88$ degenerative injuries)	Age 35–49 ($n = 63$ degenerative injuries)	Age 50–64 ($n = 24$ degenerative injuries)	Age ≥ 65 ($n = 1$ degenerative injury)
Impingement (shoulder)	28 (32%)	19 (29%)	8 (33%)	1
SLAP tear	19 (22%)	17 (27%)	2 (8%)	None
Osteoarthritis (finger)	12 (14%)	8 (13%)	4 (17%)	None
Dupuytren disease	9 (10%)	6 (10%)	3 (13%)	None
Biceps tendon rupture (LBT)	6 (7%)	4 (9%)	2 (8%)	None
Osteoarthritis elbow	6 (7%)	3 (5%)	3 (13%)	None
Osteoarthritis foot	4 (5%)	4 (6%)	None	None
Other	4 (5%)	2 (3%)	2 (8%)	None

SLAP superior labral tear from anterior to posterior; LBT long biceps tendon

tions, subacromial impingement syndrome was the most frequent diagnosis (32%), followed by chronic SLAP-lesions (superior labral tear from anterior to posterior, 22%), osteoarthritis of the fingers (14%), and Dupuytren's contracture (10%) (Table 15.2, Fig. 15.1). The subacromial impingement syndrome (Table 15.1, 10% of all patients) was found to occur twice as frequently compared to younger athletes in existing literature [4]. This is most likely influenced by the overall higher prevalence of impingement syndrome in older patients [13]. However, osteoarthritis of the finger has previously described as representing only 3% of all finger injuries in younger rock climbers [4] (Figs. 15.1 and 15.2). While this condition is the seventh leading cause of complaints in older rock climbers overall, it represents the third leading diagnosis among the degenerative conditions (Table 15.2). Our find-

ings are in agreement with a recently published study on long-term radiographic adaptations of the fingers to the stress of long-time high-level and recreational rock climbing [14].

15.4 Conclusions

While the injury distribution in older athletes is overall comparable to that of younger athletes, degenerative conditions such as impingement syndrome of the shoulder or osteoarthritis of the fingers are common injuries in older athletes. We therefore recommend careful, sport-specific medical supervision and evidence-based education under consideration of age-specific factors to prevent injuries among all age groups and to decrease the number of degenerative injuries in older rock climbers.

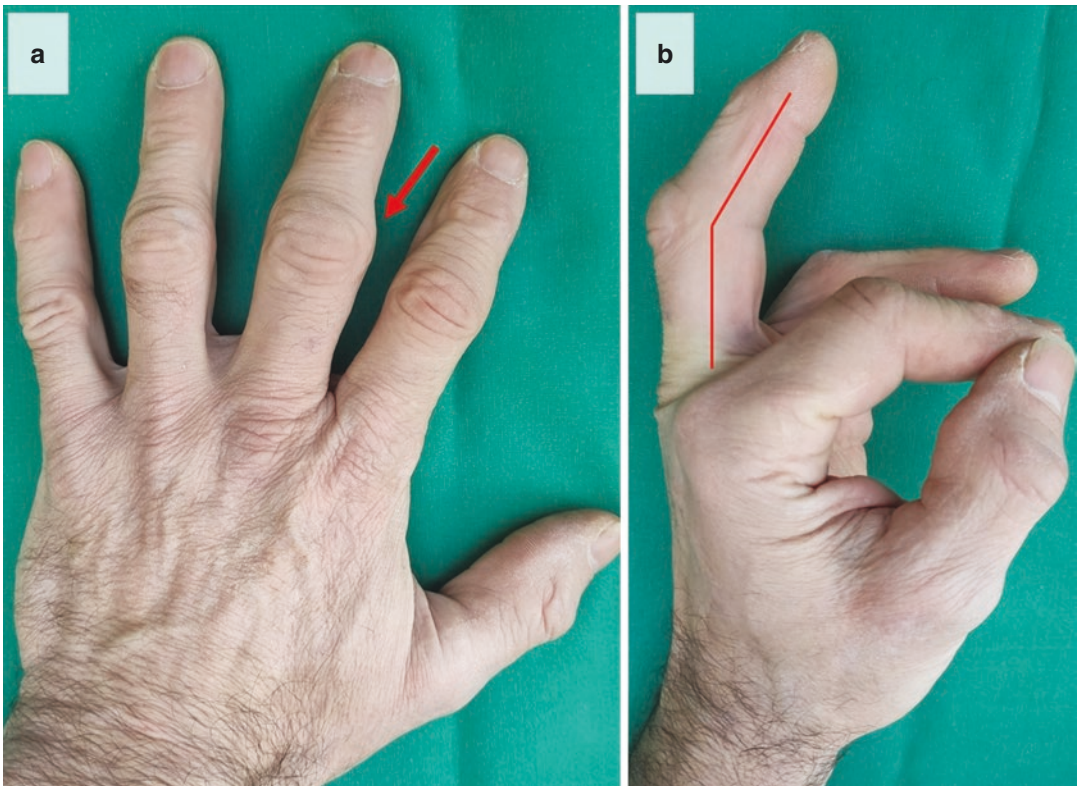
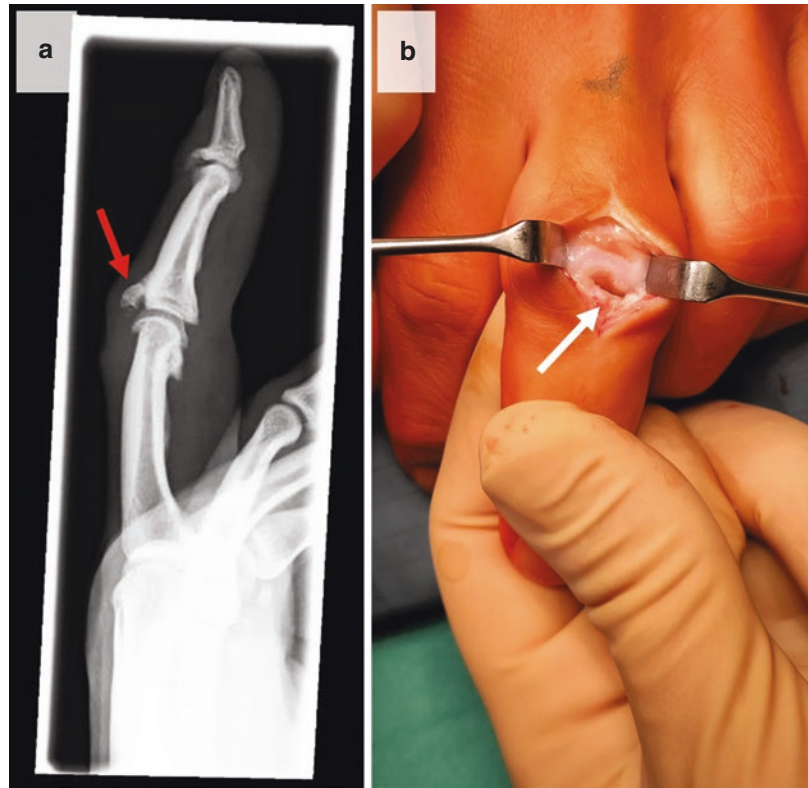


Fig. 15.1 (a) Left hand of a long-time senior rock climber (experience, approximately 40 years; UIAA level, X) with distinct distension of the PIP (red arrow) and DIP

joints. (b) In addition, there is a clear extension deficit of the fingers, here approximately 40° extension deficit of the Dig. III

Fig. 15.2 (a) X-ray of a 38-year-old male rock climber with severe bone spurs (osteophytes) at the dorsal aspect of the PIP joint (see arrow). A similar finding can be noticed in the DIP joint area. (b) Surgical treatment with removal of bone spurs around the PIP joint. The arrow marks the site of the resected osteophyte



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